BIO326 – Metagenomic Assembly and Polishing…

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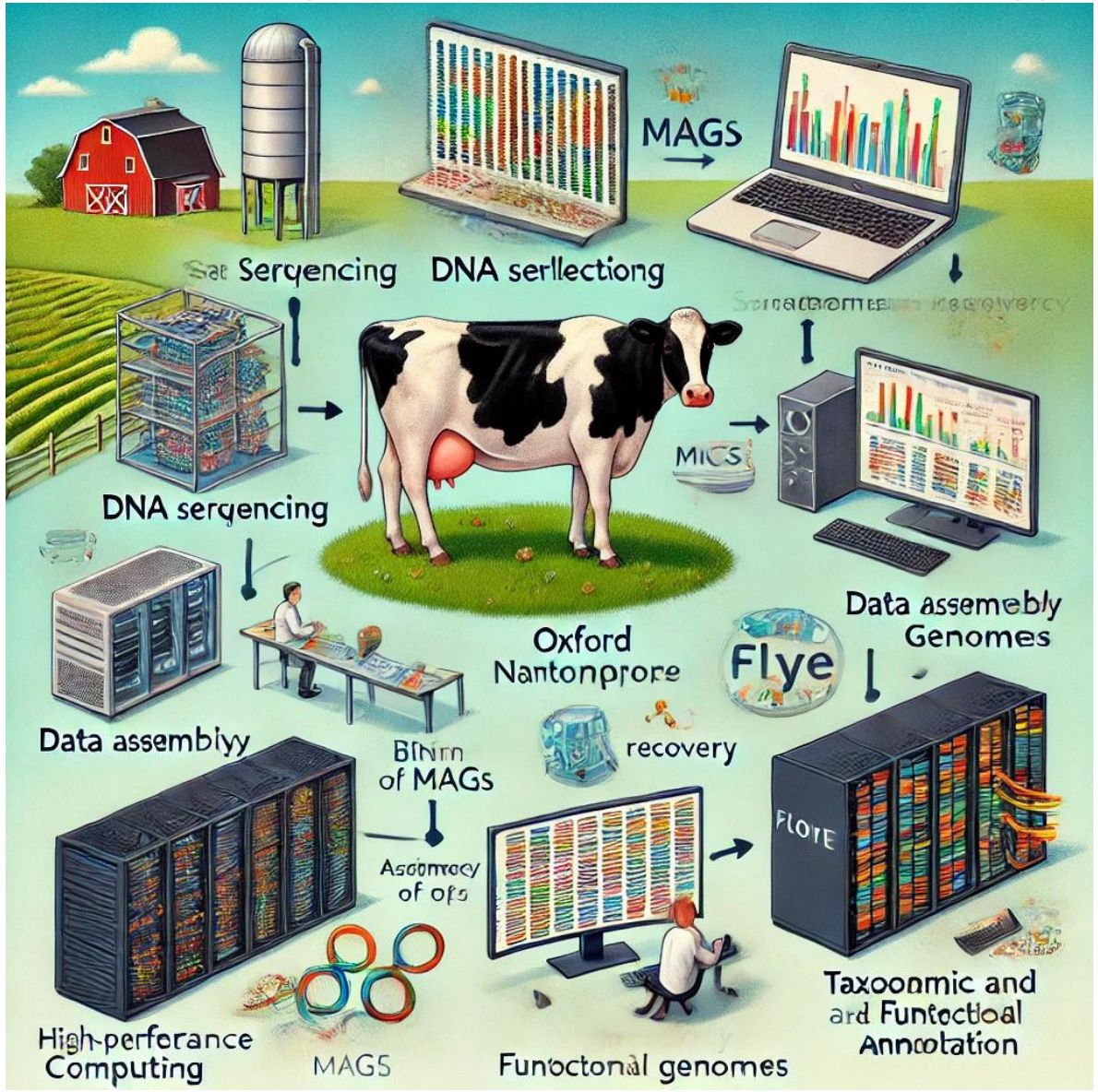
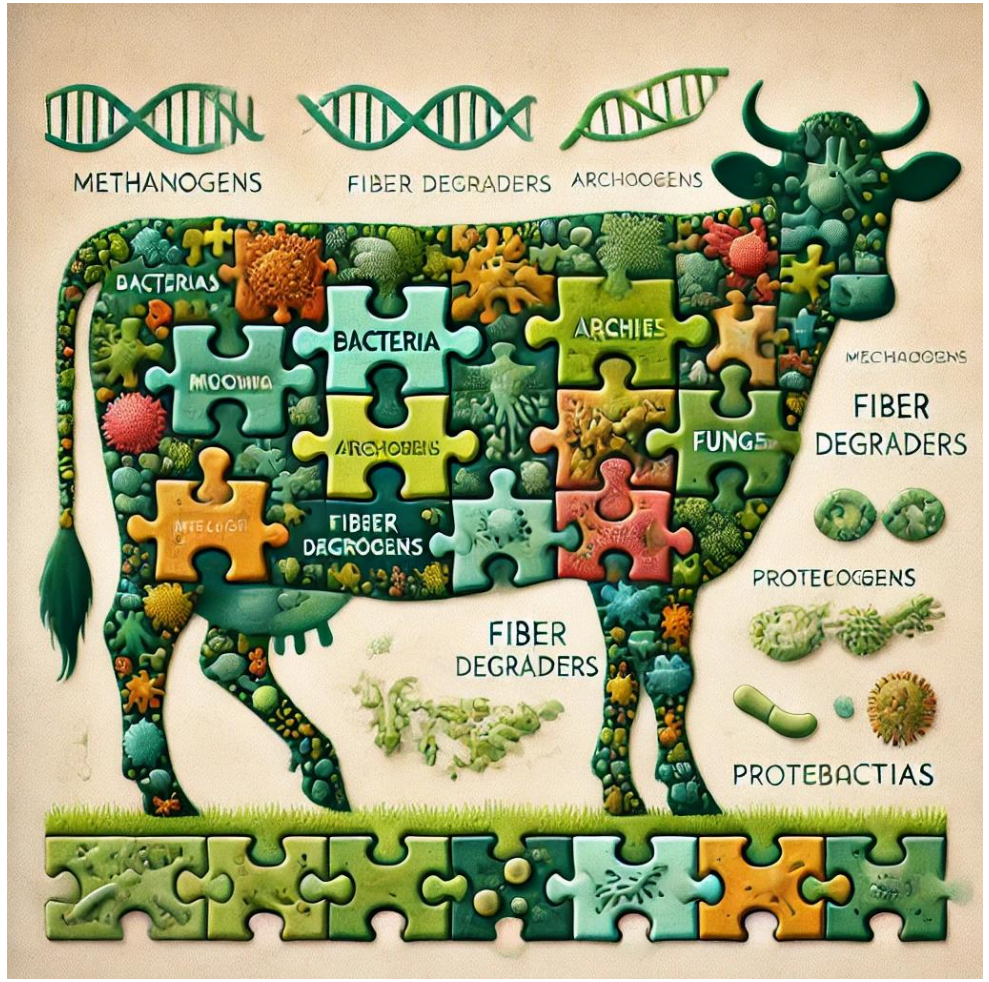
The omics give a faceted view of biology





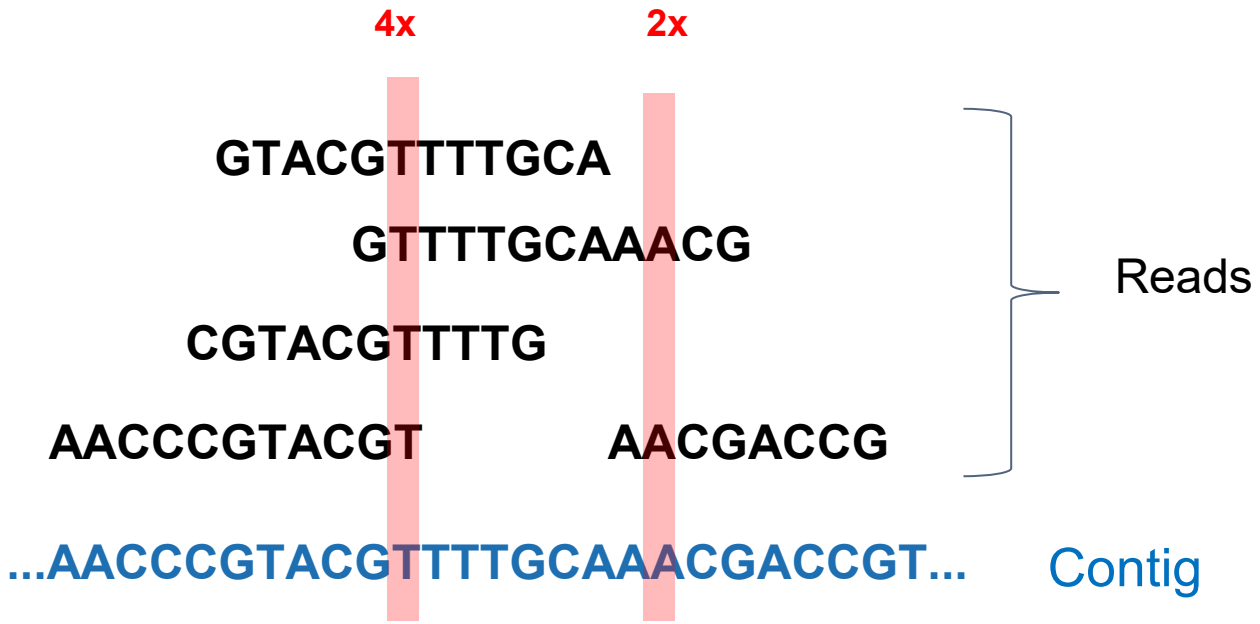
Coverage of sequencing

Definition:

- The average number of times any given nucleotide in the original genome has been sequenced

In theory, for one genome:

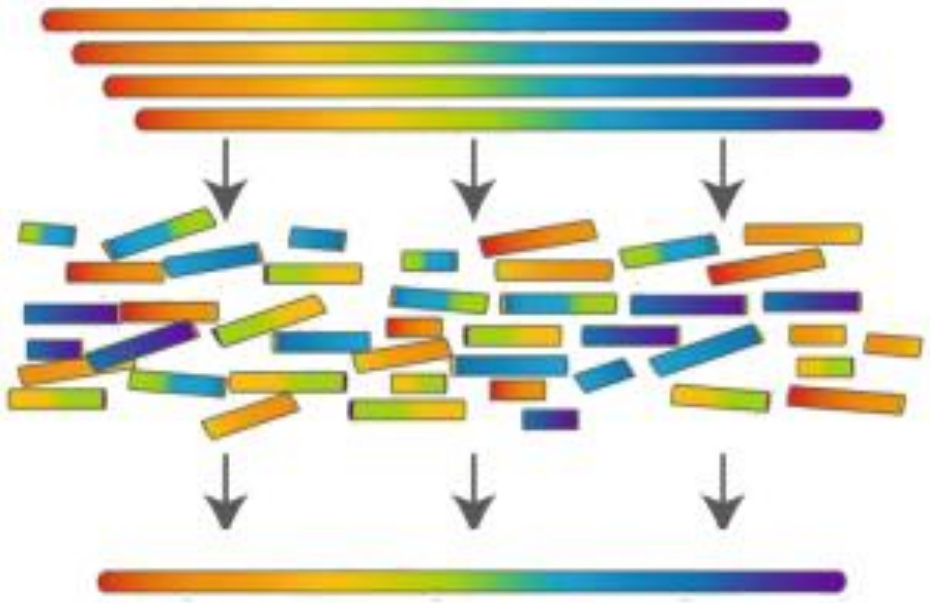
- Assumed uniformity of sequencing
- i.e. 30x coverage means the genome has been sequenced 30 times.



More about this:

- D. Sims, et al. Sequencing depth and coverage: key considerations in genomic analyses. Nat Rev Genet 15, 121–132 (2014).

Reads assembly



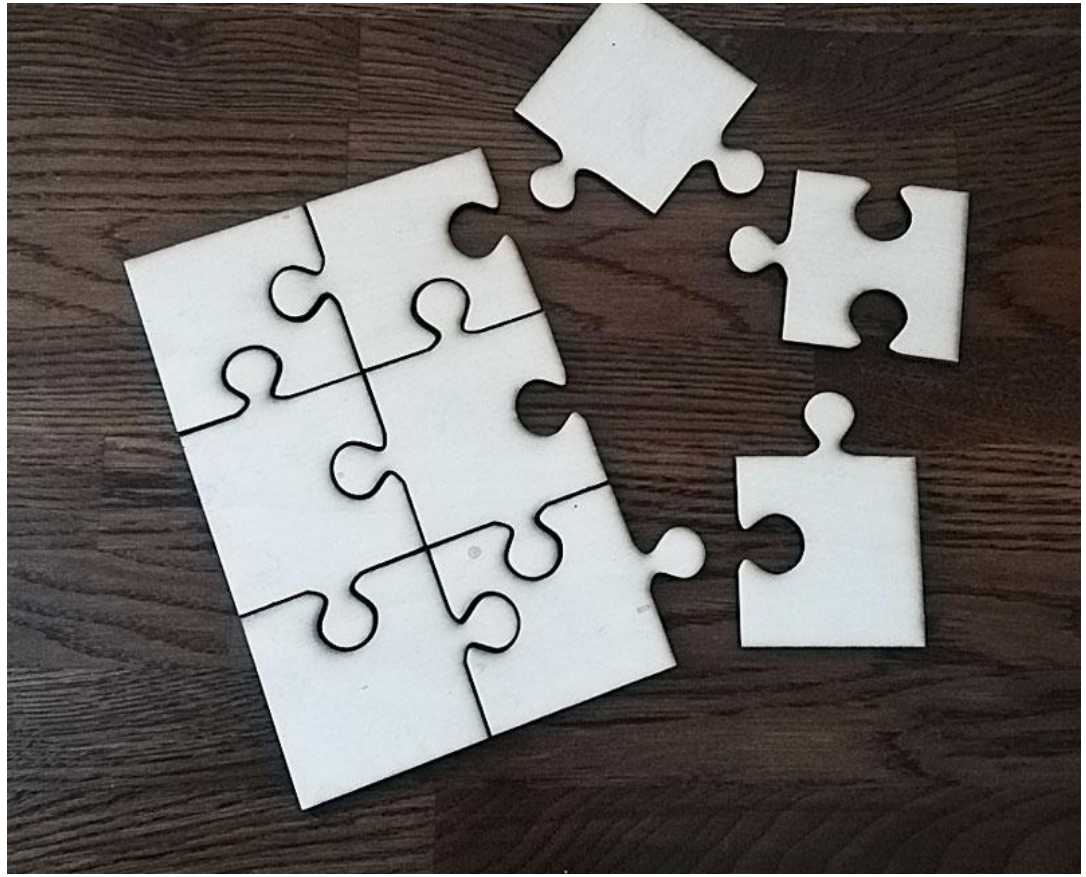
Sequence Assembly

ASSEMBLE the sequencing data into *contiguous reads*, or **CONTIGS**

<u>*Or in other words:*</u>

Resolve a puzzle and reconstruct a long DNA sequence (genome) from smaller fragments

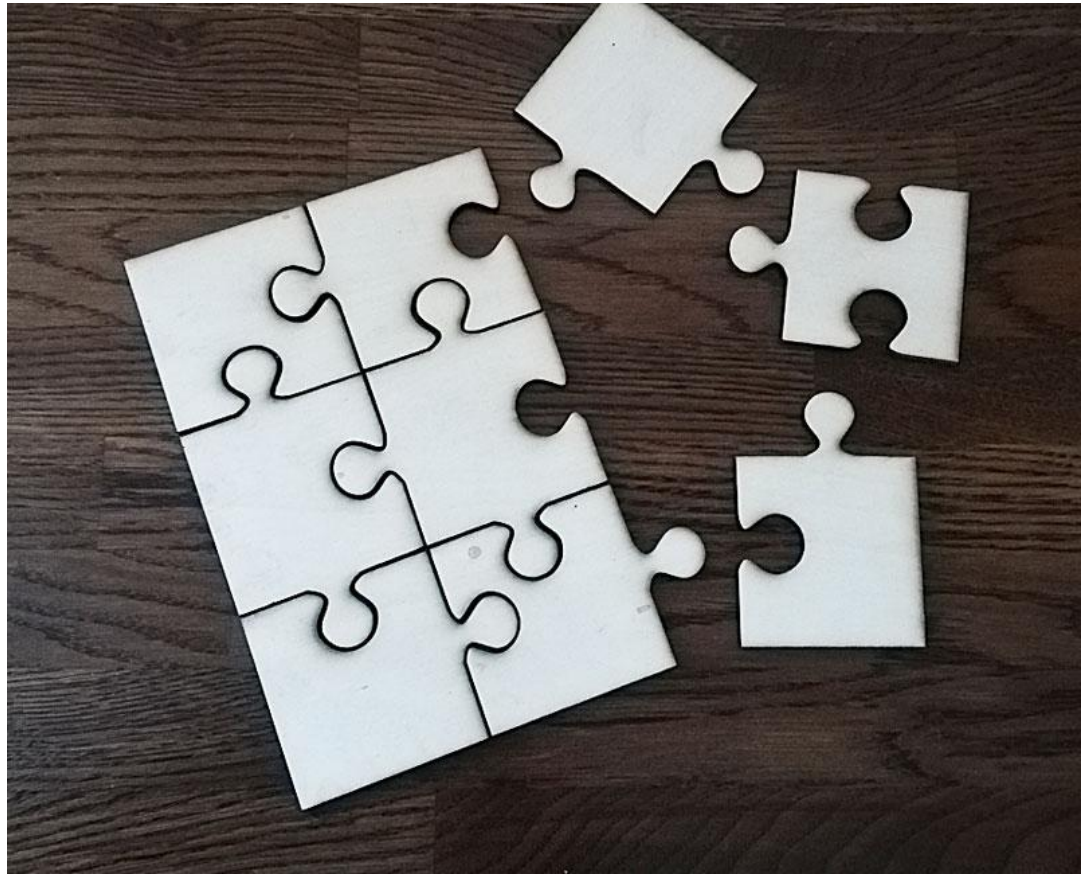
Bigger pieces (longer reads) makes the puzzle easier



Slide by Live H Hagen.

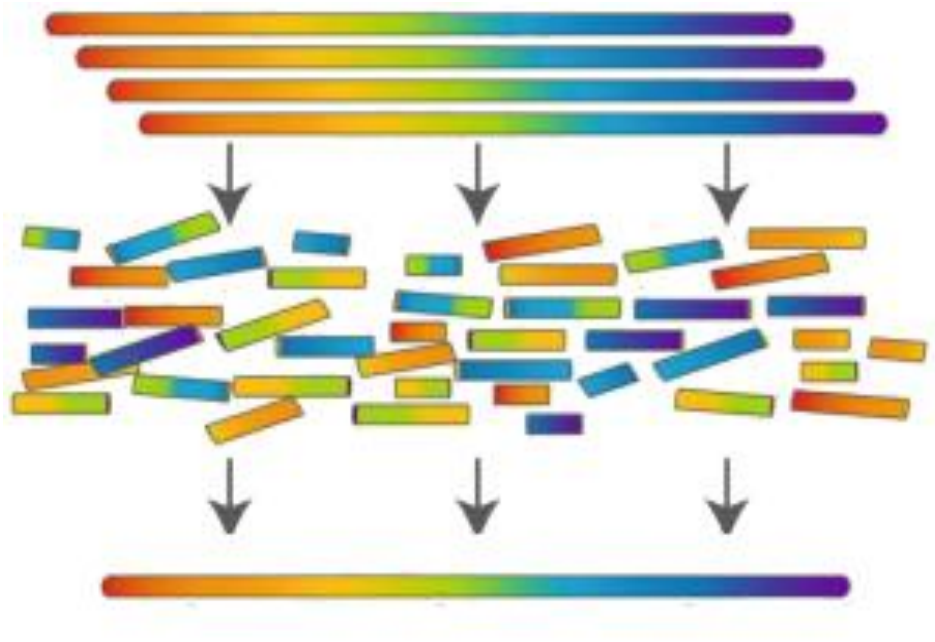
The Assembly recipe:

1. Find all **overlaps** between reads
2. Build a **graph** (find read connections)
3. **Simplify** the graph
4. Find a sensible path in the graph to generate a **consensus**



Slide by Live H Hagen.

Assembly - expectation



Genome copies

Reads

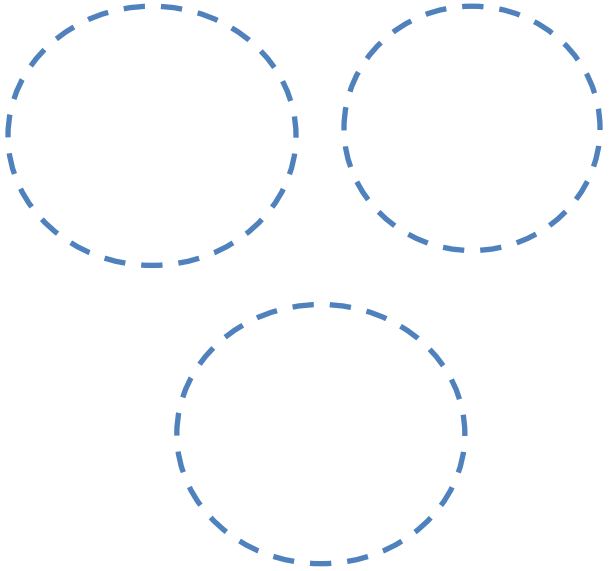
Reconstructed genome

Picture adapted from:
Commins, et al. (2009) Biological Procedures Online

Sequencing a bacterial genome

How many fragments?
or
How much coverage do we need???

Fragmentation and sequence the fragments



100 bp 500 bp

The Poisson distribution!!!!

"*The probability that a base is not sequenced is* $\mathbf{P_0 = e^{-m}}$"

m=1 equal 1 x Coverage
$P_0 = e^{-1}$=0.37 or 37 % of genome not sequenced

What does happen with 5x Coverage?

$P_0 = e^{-5}$=0.0067 or 0.67 % of genome not sequenced

And if we have 10x ???
$P_0 = e^{-10}$=4.539993e-05 or 0.000045 % of genome not sequenced
~99.995 % of the genome complete

DOI: 10.1126/science.7542800

Sequencing a bacterial genome

$C=nl/L$
C: Coverage
n: number of reads
l: length of the read
L: length of the genome

How many reads with Illumina (~100 bp) to obtain 10x coverage for a 1Mb genome?

$10x=n(100\ bp)/1x10^6$
$n=10(1x10^6)/100$
$n=1x10^5$ reads

Error rate ~ 1%

How many reads with MiniON (~50 kb) to obtain 10x coverage for a 1Mb genome?

$10x=n(5x10^4\ bp)/1x10^6$
$n=10(1x10^6)/\ 5x10^4$
n=200 reads

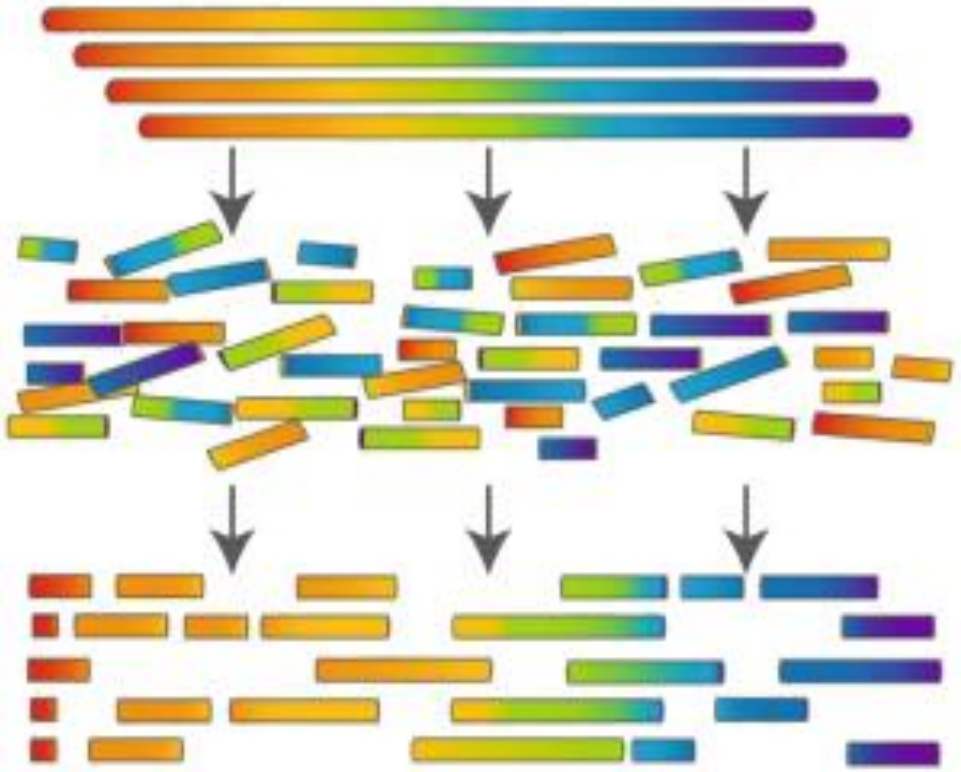
Error rate ~ 7%

Assembly - Reality

Genome copies

Reads

Contigs



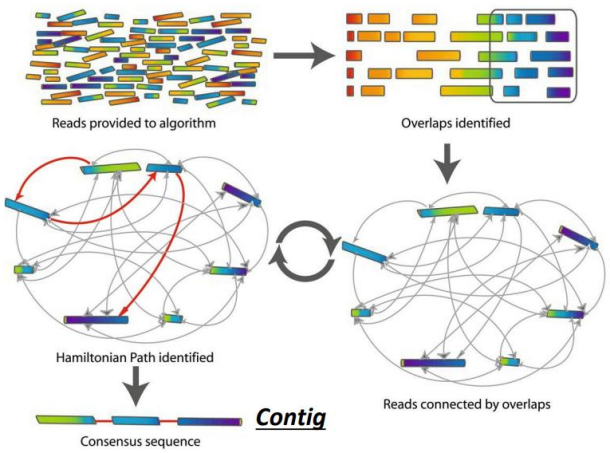
Picture adapted from:
Commins, et al. (2009) Biological Procedures Online

Two main types of assemblers

Overlap graphs (or Overlap-Layout-consensus, OLC)

OLC is composed of three main steps:

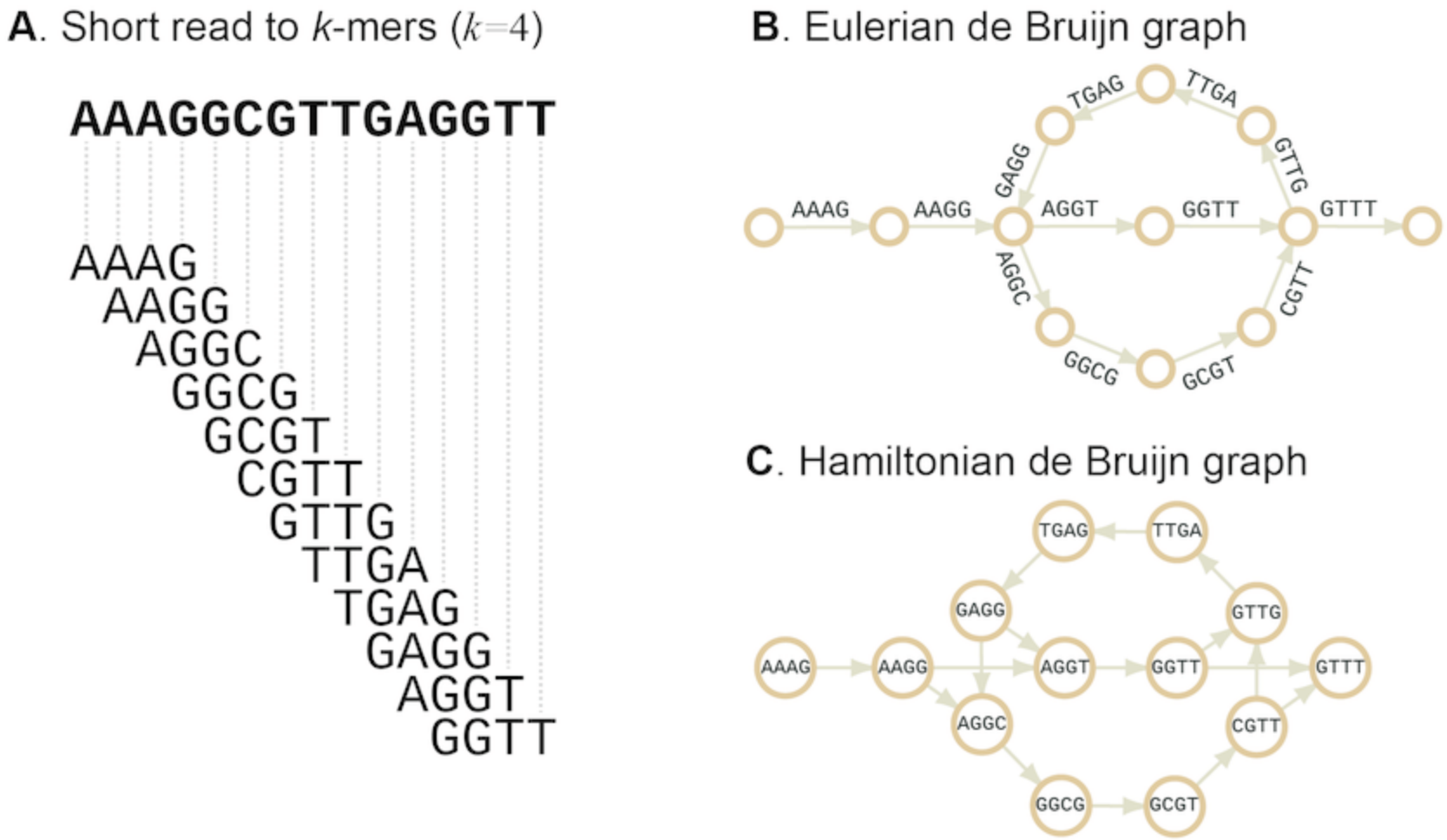
1. Computing the overlaps between the reads (blast-like alignment or k-mer)
2. Laying out the overlap information on a graph – try to simplify
3. Generate a consensus from the alignment



De Bruijn assembly:

- The (K-1)-mers are the nodes
- The prefixes/suffixes are edges
- High coverage needed
- Execution time depends on the length of the genome (O(G))
- Space depends on the k-mer composition

De Bruijn graph and k-mers



Picture adapted from:
Sohn, et al. (2016) Briefings in Bioinformatics

De Bruijn graphs – **simplified**

Let's choose a k-mer of 4
K=4

Sequence/read 1

```
AAAGGCGTTGAG

AAAG
 AAGG
  AGGC
   GGCG
    GCGT
     CGTT
      GTTG
       TTGA
        TGAG
```

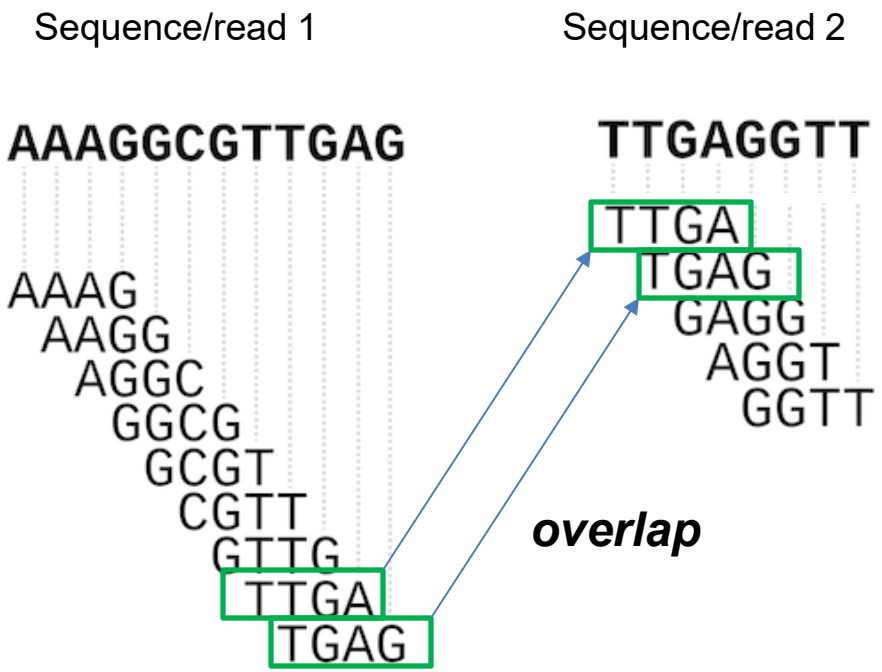
Sequence/read 2

```
TTGAGGTT

TTGA
 TGAG
  GAGG
   AGGT
    GGTT
```

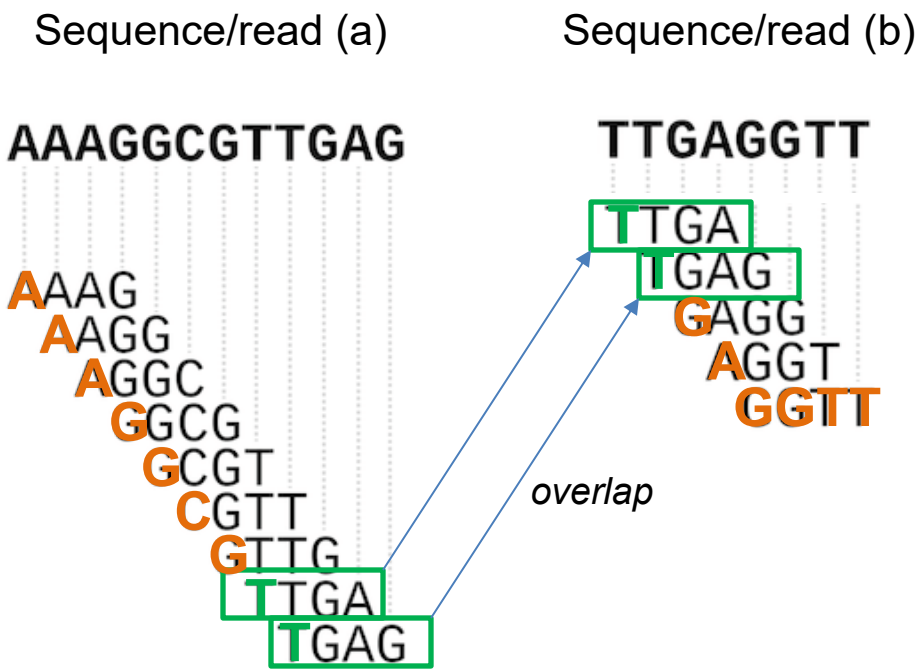
De Bruijn graphs – **simplified**

K=4



De Bruijn graphs – **simplified**

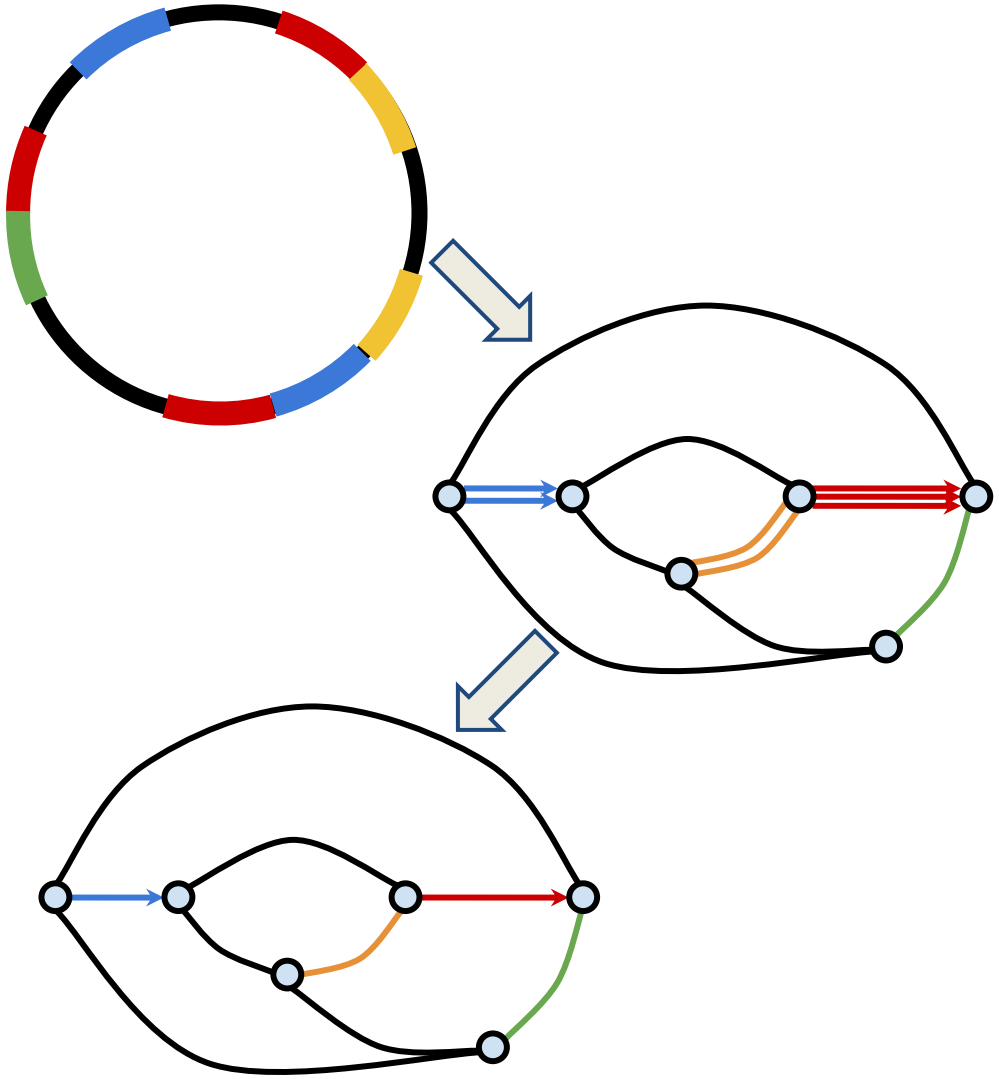
K=4



Assemblers usually use ***k-mer values*** *> 31*

AAAGGCGTTGAGGTT *Contig*

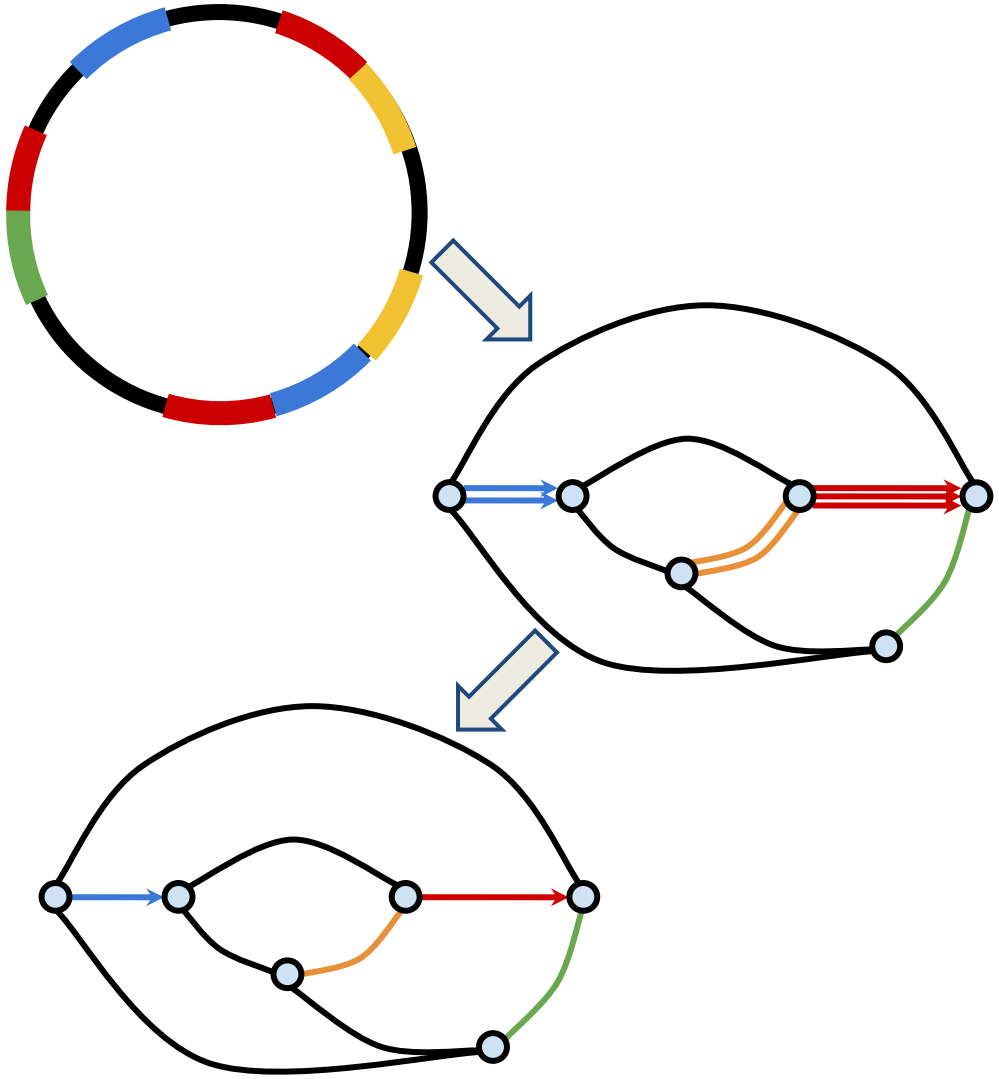
Repetitive regions are collapsed



k = k-mer size
l = length of repetitive region

If *l* > *k* the repetitive region will be collapsed in the de Bruijn graph

Is a greater *k* a solution?



k = k-mer size
l = length of repetitive region

If *l* > *k* the repetitive region will be collapsed in the de Bruijn graph

- Small *k* values save space and time, but results in a dense graph and short contigs
- Large *k* values help crossing the repetitions but is constrained by coverage and ultimately the read length

Assemblers

- Canu
- Raven
- Miniasm/minipolish

} Overlap-Layout-Consensus (OLC) assembly

- Flye
- Redbean

} Modified/fuzzy *de Bruijn* assembly / Repeat graph

- Shasta → Alignment of overlapping reads
- Unicycler → Hybrid assembly (long reads + short reads)

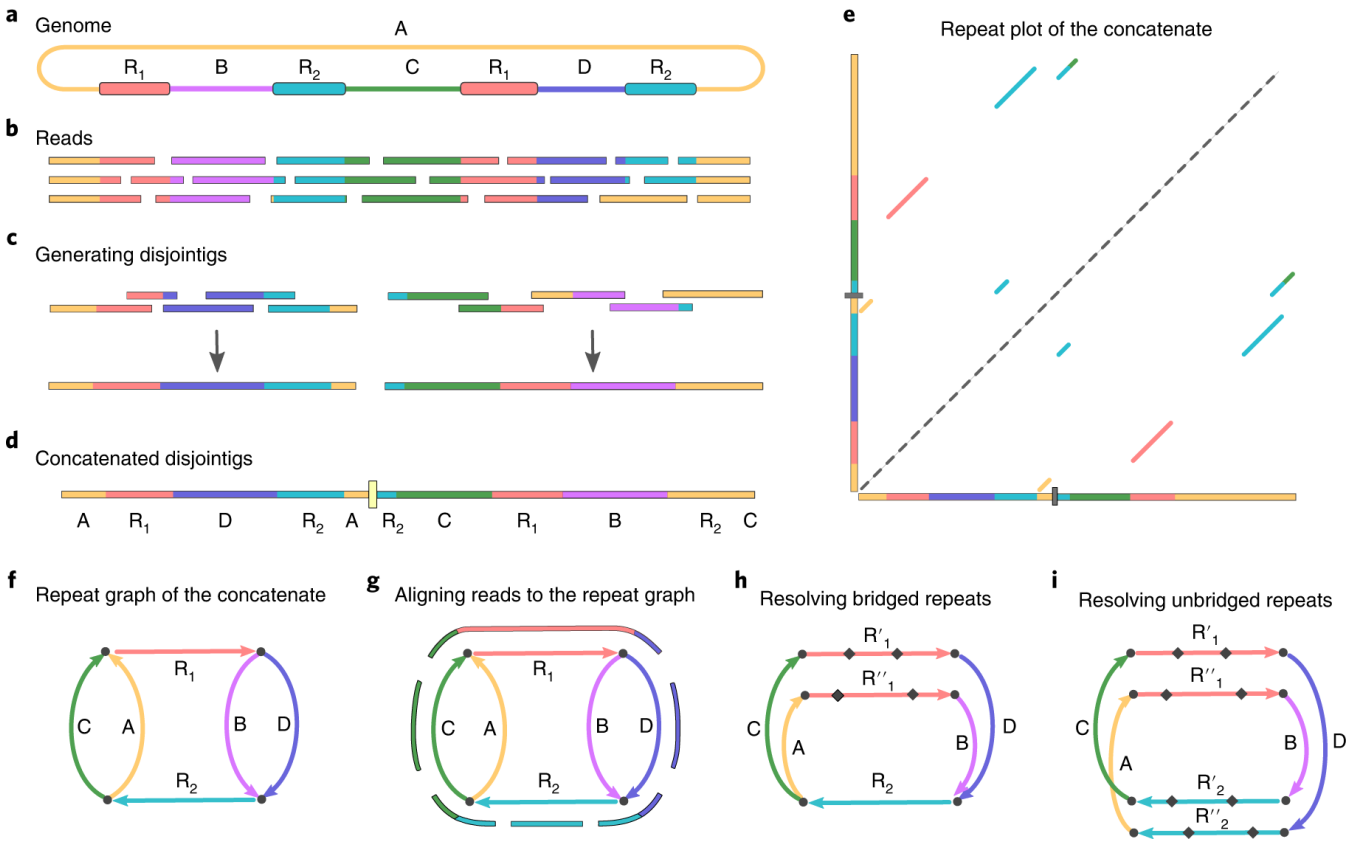
metaFly

de Bruijn graphs relies on exact k-mer matches
- But long reads have too many errors!

Solution: Repeat graphs works with approx. matches
- Tolerates errors

The Flye pipeline use repeat graphs:

1. Makes a repeat graph
2. Aligns the reads to the graph
3. Resolves the repeats



Key features

1.Graph-based Assembly:

2.Efficient Handling of Long Reads

3.Iterative Assembly Process

4.High-Performance Algorithm

- Flye uses a **minimizer-based indexing** to efficiently identify overlaps between long reads, reducing computational load and memory usage

minimizer-based indexing

- 1. Splitting Sequences into k-mers:For example,
  - from the sequence ATGCGA, with k=3
    - the k-mers are: ATG, TGC, GCG, and CGA.
- 2. Sliding Window to Find Minimizers:
  - For each k-mer, a smaller window of size w is defined, containing overlapping subsets of k-mers. Within this window, the lexicographically smallest k-mer (or based on a numerical hash function) is chosen as the minimizer.
- Lexicographical order is like dictionary order: A < C < G < T.

- Sequence:
- ATGCGT
- k-mer size k=3,
- window size W=4k-mers in window:

ATG, TGC, GCG, and CGA

- Minimizer for the window: ATG (smallest k-mer in lexicographical order).

Evaluating the assembly

Genome assembly results:

- **contig size and number of contigs produced**
- **N50 and N90**

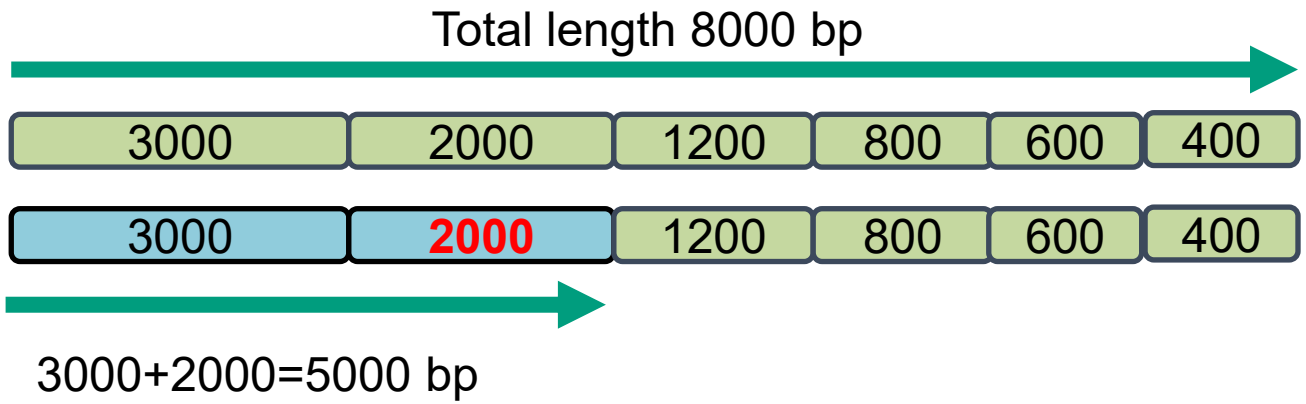
Post – assembly quality check:

- **Coverage**
- **GC Content**
- **Genome annotation (**including annotation of microRNA, tRNA, rRNA, and other ncRNA)

Basic stats

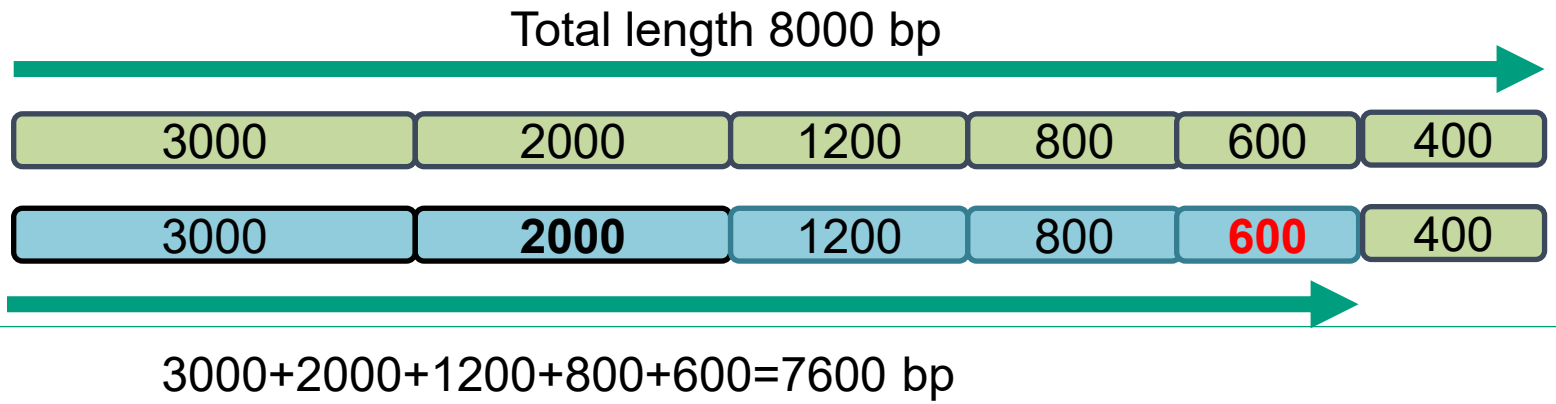
N50 the length of the shortest contig such that the sum of contigs of equal length or longer is at least 50% of the total length of all contigs.

50% of
8000=4000



N90 = the length of the shortest contig such that the sum of contigs of equal length or longer is at least 90% of the total length of all contigs.

90% of
8000=7200



Polish assembly:

🧬 Polishing Oxford Nanopore Data with Medaka: A Summary

Medaka is a polishing tool developed by **Oxford Nanopore Technologies (ONT)** specifically for improving the **base-level accuracy** of genome assemblies generated from **long-read Nanopore sequencing data**.

✅ Why Polishing is Needed

Oxford Nanopore reads are long but have **higher base error rates** (~5–15%) compared to short reads (e.g., Illumina). After assembling the genome (e.g., with Flye, Canu, or Shasta), polishing is crucial to:

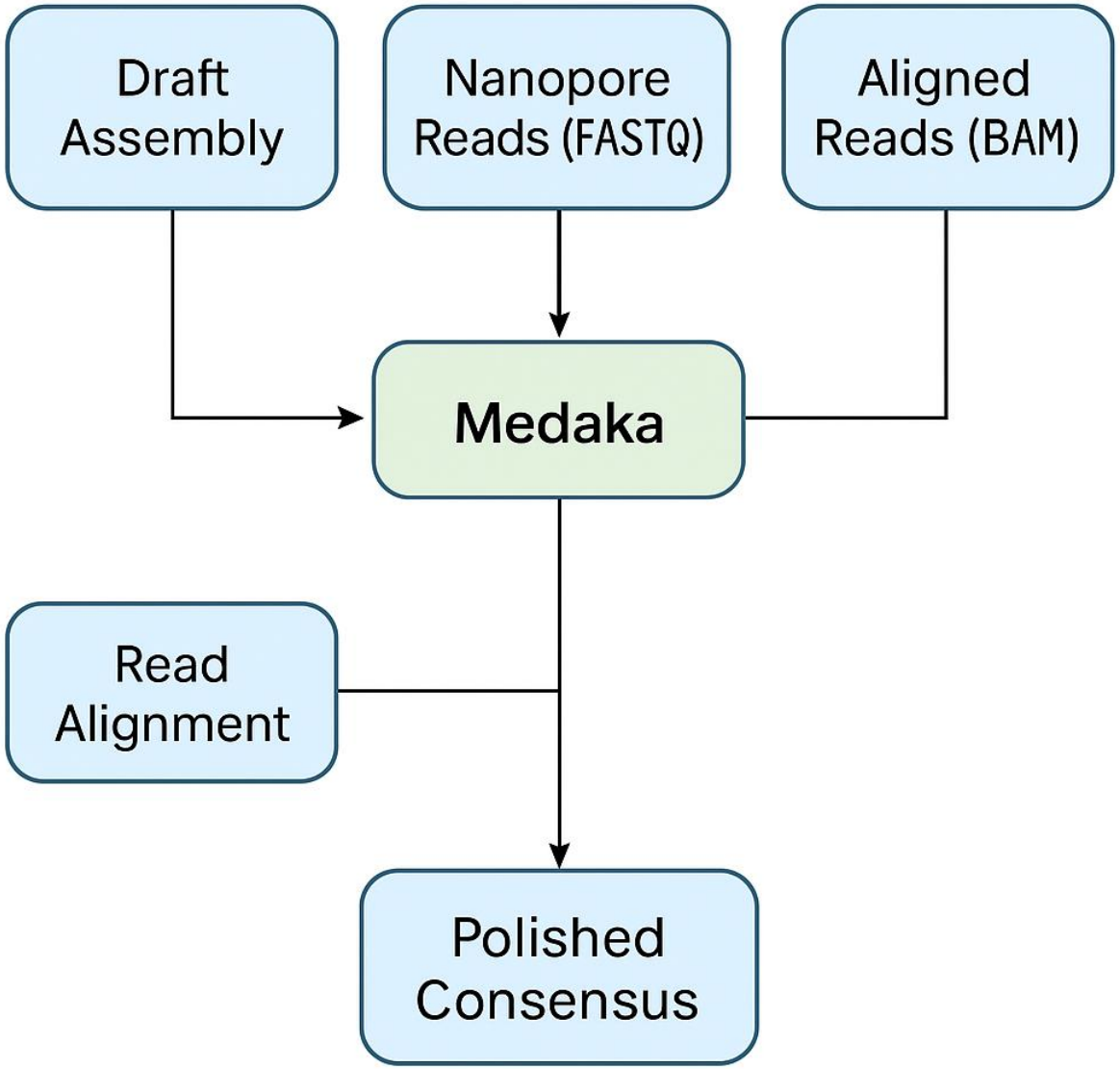
- Correct base errors (substitutions, indels)
- Improve the consensus sequence
- Enhance gene predictions and downstream analyses

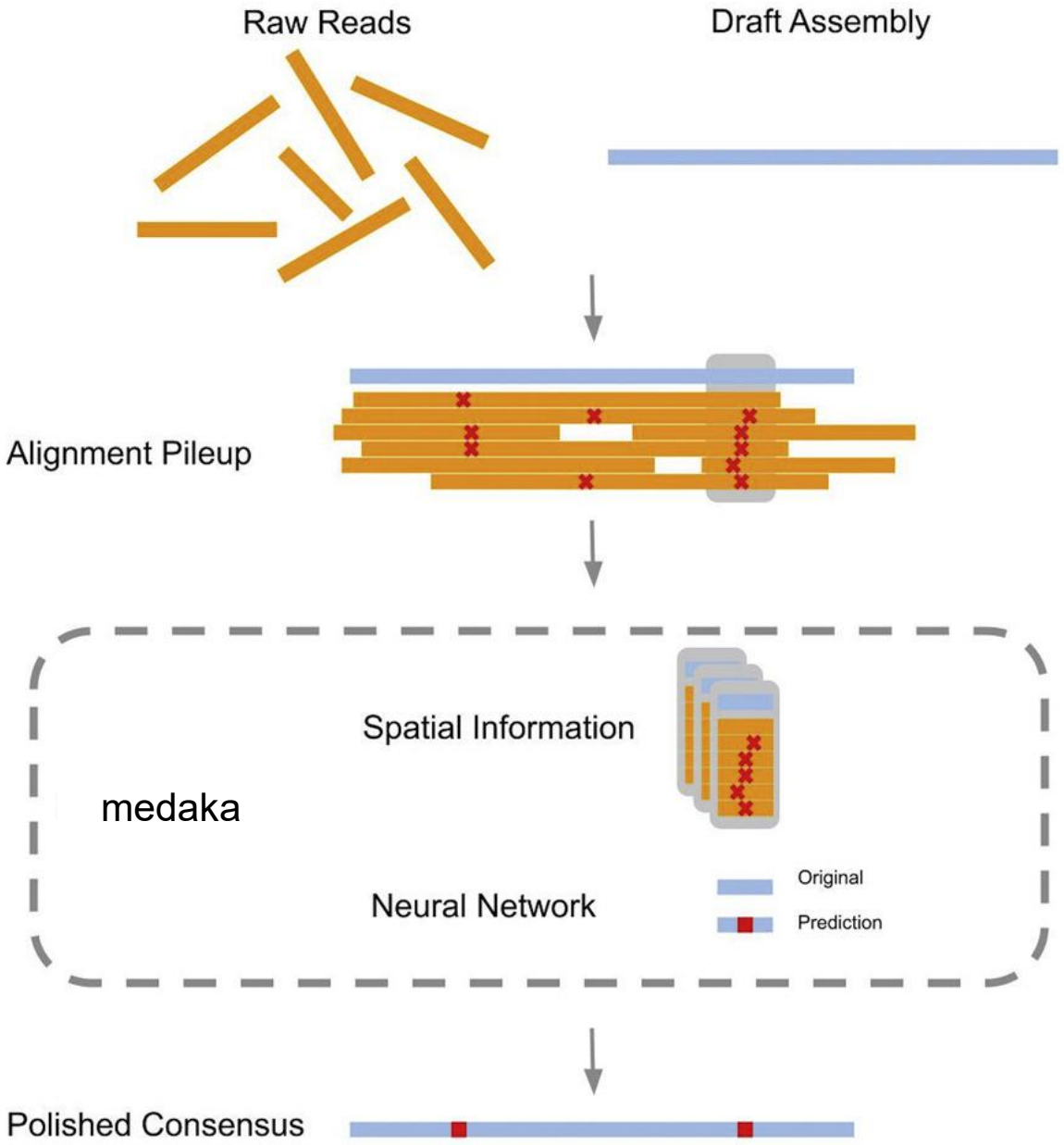
Inside MEDAKA

⚙️ How Medaka Works

1. **Input Requirements:**
   - A draft genome assembly (FASTA file)
   - Aligned Nanopore reads (BAM file, sorted and indexed)
   - Basecalled Nanopore reads (FASTQ)
   - A model trained for your specific basecaller version and read type
2. **Steps in the Pipeline:**
   1. **Read Alignment:**
      - Long reads are aligned to the draft assembly using **minimap2**.
   2. **Neural Network Inference:**
      - Medaka uses a **pre-trained neural network** to predict the most likely true sequence from the pileup of reads at each base position.
   3. **Consensus Correction:**
      - It corrects errors in the draft by applying the model's prediction, producing a **polished consensus FASTA** file.

How it works?





Adapted from: Zhang, Yifan et al.
iScience, Volume 23, Issue 5, 101128